The Duval Pentagon—A New Complementary Tool for the Interpretation of Dissolved Gas Analysis in Transformers

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The article presents Duval Pentagons as new complementary tools for the interpretation of dissolved gas analysis in mineral oil–filled electrical equipment.

Introduction

Several methods are available for the interpretation of dissolved gas analysis (DGA) results in oil-filled electrical equipment. These methods use ratios of the five main hydrocarbon gases formed, namely H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2, either 2-gas ratios, e.g., in IEEE [1] and IEC [2]; 3-gas ratios, in Duval Triangles 1 to 7 [3]; or 4-gas ratios in [4]. In this article a new method is presented using 5-gas ratios in a "pentagon" representation applicable to mineral oil–filled equipment. Carbon oxides CO and CO_2 are examined separately as in other methods.

The Duval Pentagon

In the new Duval Pentagon representation, the relative percentages of the five main hydrocarbon gases analyzed by DGA are first calculated. For instance the relative percentage of H_2 = (ppm of H_2)/(ppm of $H_2 + CH_4 + C_2H_6 + C_2H_4 + C_2H_2$).

An example of the Duval Pentagon representation and calculations used is illustrated in Figure 1. Each summit of the pentagon corresponds to one gas, e.g., H_2. The relative percentage of H_2 is plotted on the axis between the pentagon center (0% H_2) and the pentagon summit for H_2 (100% H_2). The same is done for each of the other 4 gases.

In the example in Figure 1, the DGA results were H_2 = 31 ppm, C_2H_6 = 130 ppm, CH_4 = 192 ppm, C_2H_4 = 31 ppm, and C_2H_2 = 0 ppm. The relative percentage of each gas (8, 34, 50, 8, and 0%, respectively) was plotted on its corresponding gas axis, providing five different points represented as red squares. The center ("centroid") of the irregular polygon drawn from these five points was then calculated mathematically as indicated below, providing a sixth point represented by a blue square in Figure 1. This last point represents the DGA results of this example in the pentagon configuration.

Calculation of the Centroid Coordinates

To calculate the coordinates of the centroid of the five red points in Figure 1, a horizontal axis (x) and a vertical axis (y) intersecting at the pentagon center are first added, as shown in Figure 1.

The (x_i, y_i) coordinates of each of the five points are then calculated. For instance for the point on the C_2H_6 axis, the angle α between the C_2H_6 axis and the x axis is 18 degrees. So its coordinates are $x_1 = 34\ (\%) \times \cos\alpha = -32.3$, and $y_1 = 34 \times \cos(90 - \alpha) = 10.5$. The same is done for the other four points. In this example the (x_i, y_i) coordinates for H_2, CH_4, C_2H_4, and C_2H_2 are thus (0, 8.1), (−29.4, −40.5), (4.8, −6.5), and (0, 0), respectively.

The (x, y) coordinates of the centroid of these five points are then calculated using the equations indicated in [5]:

$$C_x = \frac{1}{6A}\sum_{i-0}^{n-1}(x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

$$C_y = \frac{1}{6A}\sum(y_i + y_{i+1})(x_i y_{i+1} - x_{i+1} y_i),$$

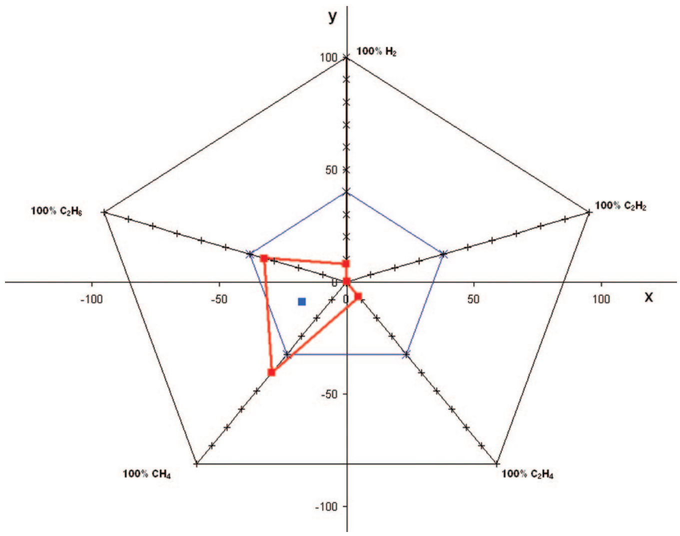

Figure 1. Example of Duval Pentagon representation.

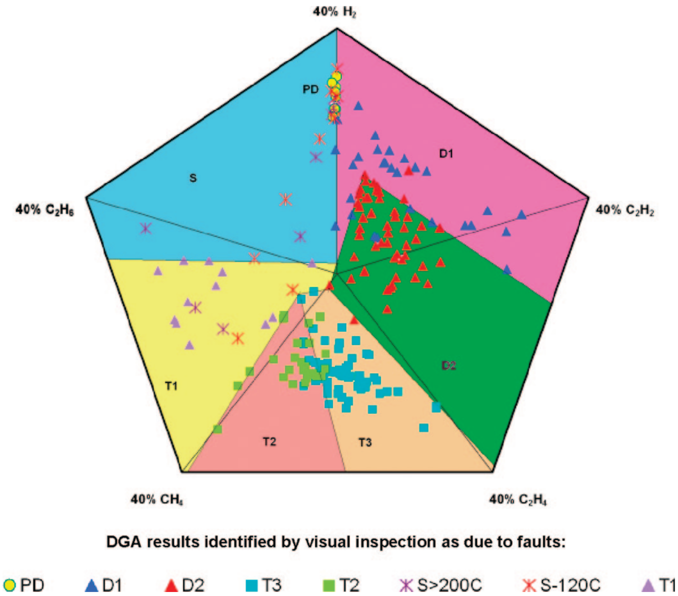

Figure 2. The Duval Pentagon 1 for the six "basic" faults, PD, D1, D2, T3, T2, and T1, and stray gassing of mineral oil S (see text for meanings of the abbreviations).

where x_i and y_i are the coordinates of the five points, C_x and C_y the (x, y) coordinates of the centroid, and A the surface of the polygon:

$$A = \frac{1}{2}\sum_{i-0}^{n-1}(x_i y_{i+1} - x_{i+1} y_i).$$

In the example shown in Figure 1, the (x, y) coordinates of the centroid are (−17.3, −9.1). It may be noted that the center of an irregular polygon may also be calculated mathematically as its "center of mass," by computing the average of its (x_i) and (y_i) coordinates, but it was preferred here to calculate it as its "centroid."

The order of gases at the five summits of the pentagon corresponds to the increasing energy required to produce these gases in transformers, from H_2 to C_2H_2, counterclockwise in Figure 1, as in the case of Duval Triangles 1, 4, and 5 [3]. This order was confirmed to provide the best fit in terms of identification of faults in the pentagon representation.

The (x, y) coordinates of the 100% summits of the pentagon for H_2, C_2H_6, CH_4, C_2H_4, and C_2H_2 are (0, 100), (−95.1, 30.9), (−58.8, −80.9), (58.8, −80.9), and (95.1, 30.9), respectively.

Even when the relative percentage is 100% for one gas, e.g., H_2, and 0% for the other gases, the centroid will be at no more than 40% on the H_2 axis. So, in practice, for positioning DGA centroid points in the pentagon, units on each gas axis can be limited to 40%.

The (x, y) coordinates of the 40% summits of the pentagon for H_2, C_2H_6, CH_4, C_2H_4, and C_2H_2 are (0, 40), (−38, 12.4), (−23.5, −32.4), (23.5, −32.4), and (38, 12.4), respectively.

Fault Zones in the Duval Pentagon

To define fault zones in the Duval Pentagon representation, about 180 DGA results due to faults identified by visual inspection of the mineral oil–filled transformers were used to establish the centroid. The Duval Pentagon 1 is shown in Figure 2, with fault zones in colors corresponding to the six "basic" electrical and thermal faults used by IEC, IEEE, and Duval Triangle 1. These are

- PD: corona partial discharges,
- D1: low energy discharges,
- D2: high energy discharges,
- T3: thermal faults >700°C,
- T2: thermal faults of 300 to 700°C, and
- T1: thermal faults <300°C.

Figure 2 also shows an additional zone S for stray gassing of *mineral* oil using results of ~20 stray gassing tests at 120 and 200°C in the laboratory [6]. The individual points in color in Figure 2 correspond to the centroid points of DGA results identified by visual inspection as due to one of these faults.

Figure 3 shows Duval Pentagon 2, with fault zones in colors corresponding to the three basic electrical faults, PD, D1, and D2, and the four more precisely defined, or "advanced," thermal faults used in Duval Triangles 4 and 5 [7]:

- T3-H: thermal faults T3-H in oil only,
- C: thermal faults T3-C, T2-C, and T1-C with carbonization of paper,
- O: overheating T1-O <250°C, and
- S: stray gassing S of mineral oil at 120 and 200°C in the laboratory.

The individual points in colors in Figure 3 correspond to the centroids of DGA results identified by visual inspection as due to one of these faults, as well as to thermal faults T2-H and T1-H in oil only. Note that the "H" in T3-H, T2-H, and T1-H means "Huile" for "Oil" in French.

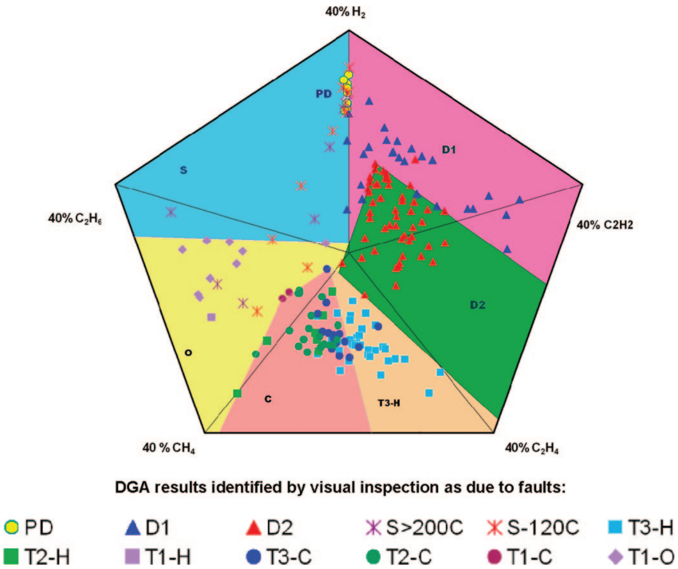

Figure 3. The Duval Pentagon 2 for the three basic electrical faults PD, D1, and D2 and the four "advanced" thermal faults T3-H, C, O, and S (see text for meanings of the abbreviations).

The individual DGA centroid coordinates and fault zones in Pentagons 1 and 2 are, in general, in good agreement, except for a few outliers, as with other diagnosis methods, which may be due to inaccurate DGA results, inaccurate inspection reports, or unidentified mixtures of faults. There is also some overlap in Pentagon 2 between faults C and faults T2-H in oil only. However, carbonization of paper was found in ~80% of DGA cases occurring in zone C.

The (x, y) coordinates of the summits of zone boundaries in Pentagons 1 and 2 are as follows:

- PD: (0, 24.5), (0, 33), (−1, 24.5), (−1, 33),
- D1: (0, 40), (38, 12), (32, −6), (4, 16), (0, 1.5),
- D2: (4, 16), (32, −6), (24, −30), (−1, −2),
- T3: (24, −30), (−1,−2), (−6,−4), (1, −32),
- T2: (1, −32), (−6, −4), (−22.5, −32),
- T1: (−22.5, −32), (−6, −4), (−1, −2), (0, 1.5), (−35, 3),
- S: (−35, 3), (0, 1.5), (0, 24.5), (0, 33), (−1, 24.5), (−1, 33), (0, 40),
- T3−H: (−24, −30), (−3.5, −3), (2.5, −32),
- C: (2.5, −32), (−3.5, −3), (−11, −8), (−21.5, −32), and
- O: (−21.5, −32), (−11,−8), (−3.5, −3), (−1, −2), (0, 1.5), (−35, 3).

Algorithm versions of Pentagons 1 and 2 are available at no cost from duvalm@ireq.ca.

Using the Pentagons

One set of DGA results from a mineral oil–filled transformer in service will provide coordinates for one centroid in one fault zone of the Pentagons, thus allowing the identification of the fault. Pentagons 1-2 may be used alone or in combination with Triangles 1-4-5 for mineral oils in order to get more information about the fault. Pentagons 1 and 2, like Triangles 4 and 5, permit distinguishing between thermal faults of lesser concern for the equipment, e.g., T3-H and S faults in oil and O, overheating from faults possibly involving carbonization of paper C, which are potentially more dangerous.

When Pentagons 1-2 and Triangles 1-4-5 indicate different types of faults for the same set of DGA results, this may be an indication of a mixture of faults in the transformer. This is because each Triangle or Pentagon representation gives more weight to some gases and therefore to one of the faults in the mixture. Comparing them may thus help identify the different faults involved in the mixture. For instance, in the case of a mixture of faults, S and T3, Triangle 4 will be more sensitive to fault S, because of H_2, and Triangle 5 to fault T3, because of C_2H_4, while the DGA point in the Pentagons will appear intermediate between these two faults, thus confirming a mixture of faults.

Conclusions

New Duval Pentagons 1 and 2 for the interpretation of DGA results in mineral oil–filled transformers and similar equipment, e.g., bushings and cables, are presented in this article, allowing the use of the five main diagnosis gases, H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2, in a single graphical representation. Although they may be used alone, they are not intended to replace Duval Triangles 1, 4, and 5 for mineral oils but rather to bring complementary information, for instance for the case of mixtures of faults. Other versions applicable to non–mineral oils (natural and synthetic esters, silicones) and corresponding to Duval Triangles 3, 6, and 7 will be developed later.

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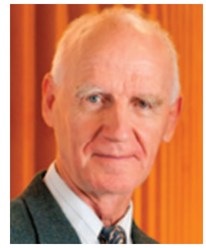

Michel Duval obtained a BSc and PhD in chemical engineering in 1966 and 1970 and has worked for IREQ (Hydro-Quebec, Canada) since 1970. In the field of DGA he is well-known for his Triangle method of interpretation, used worldwide, and for his other contributions to DGA (gas-in-oil standards, gas levels in service, on-line gas monitors). He has also been active in the field of electrical insulating oils and polymers. Duval is the convener of several CIGRE, IEC, and IEEE working groups. He

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